



BASIC OF CIVIL ENGINEERING (TCE 101)

UNIT- 2 (SYLLABUS)

Surveying:

- Introduction: Object, Principles & Types of Surveying; Site Plans, Plans & Maps; Scales & Unit of different Measurements.
- Linear Measurements: Linear Measurement by Tape, Ranging out Survey Lines and overcoming obstructions.
- Modern distance measuring equipment like Laser Distance Meter (LDM), etc.
- Angular Measurements: Introduction to Compass Surveying, Bearings. Introduction to the total station.
- Modern Tools of Surveying and Mapping: Introduction to Global Positioning System, Remote Sensing and Geographic Information System.
- Survey Lab Visit and measurement activity.

Definition of Surveying

Surveying is the art and science of determining the relative positions of various points or stations on the surface of the earth by measuring the horizontal and vertical distances, Angles and taking the details of these points and by preparing a map or plan to any scale.



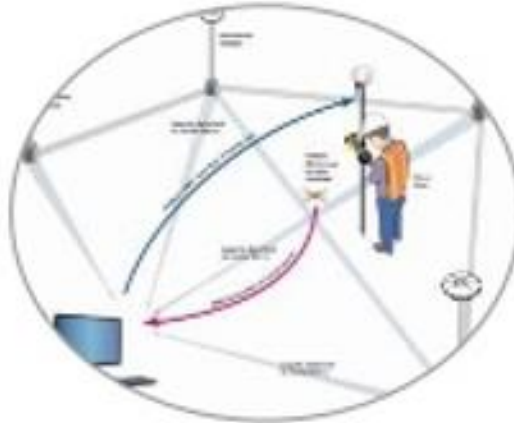
★ Measurements taken in Horizontal and Vertical planes

Objects of Surveying



To Prepare a map or plan.

To show the relative position of the objects on the surface of the earth.



To establish the boundaries of the land.



To select a suitable site for an engineering project.



Fundamental Principles of Surveying

Principle: 1

WHOLE TO PART

Always work from whole to part

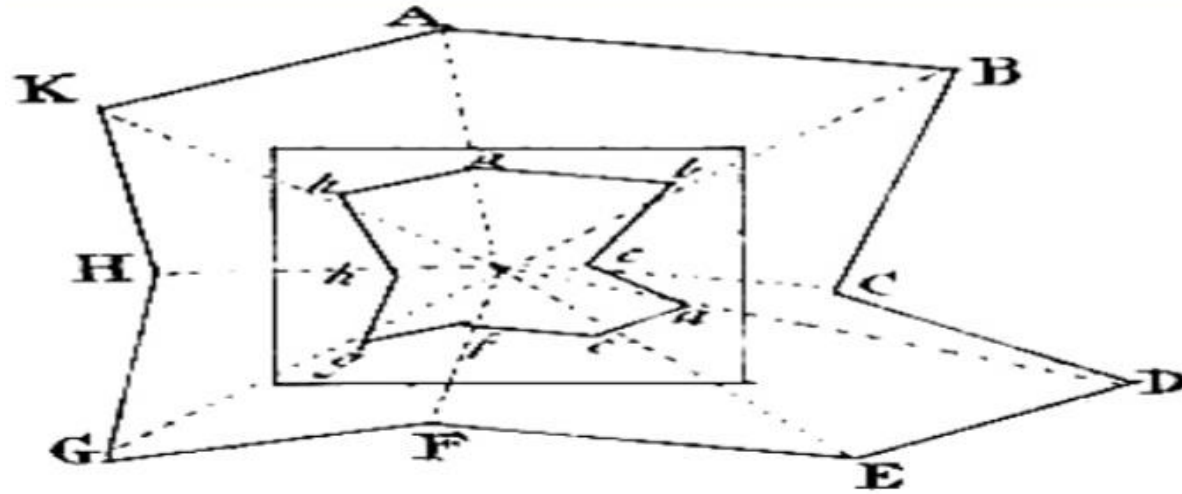
Principle: 2

LOCATE A POINT BY ATLEAST
TWO MEASUREMENTS

Locate a new station by at least 2
measurements whether linear or
angular from fixed reference
points.

Fundamental Principles of Surveying

Always work from whole to part:

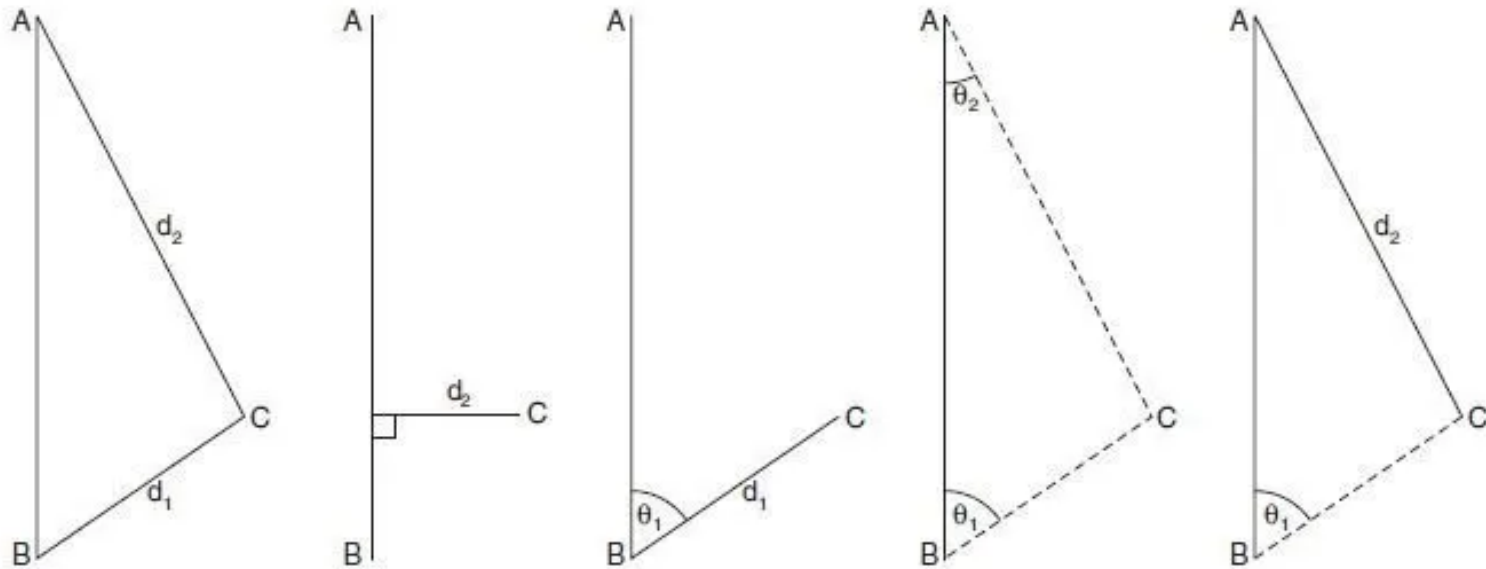


According to the first principle, the whole survey area is first enclosed by main stations (i.e.. Control stations) and main survey lines. The area is then divided into a number of divisions by forming well conditioned triangles.

The main survey lines are measured very accurately with precise survey instruments. The remaining sides of measured. The purpose of this method of working The triangle are is to control accumulation of errors. During measurement, if there is any error, then it will not affect the whole work, but if the reverse process is followed then the minor error in measurement will be magnified.

Fundamental Principles of Surveying

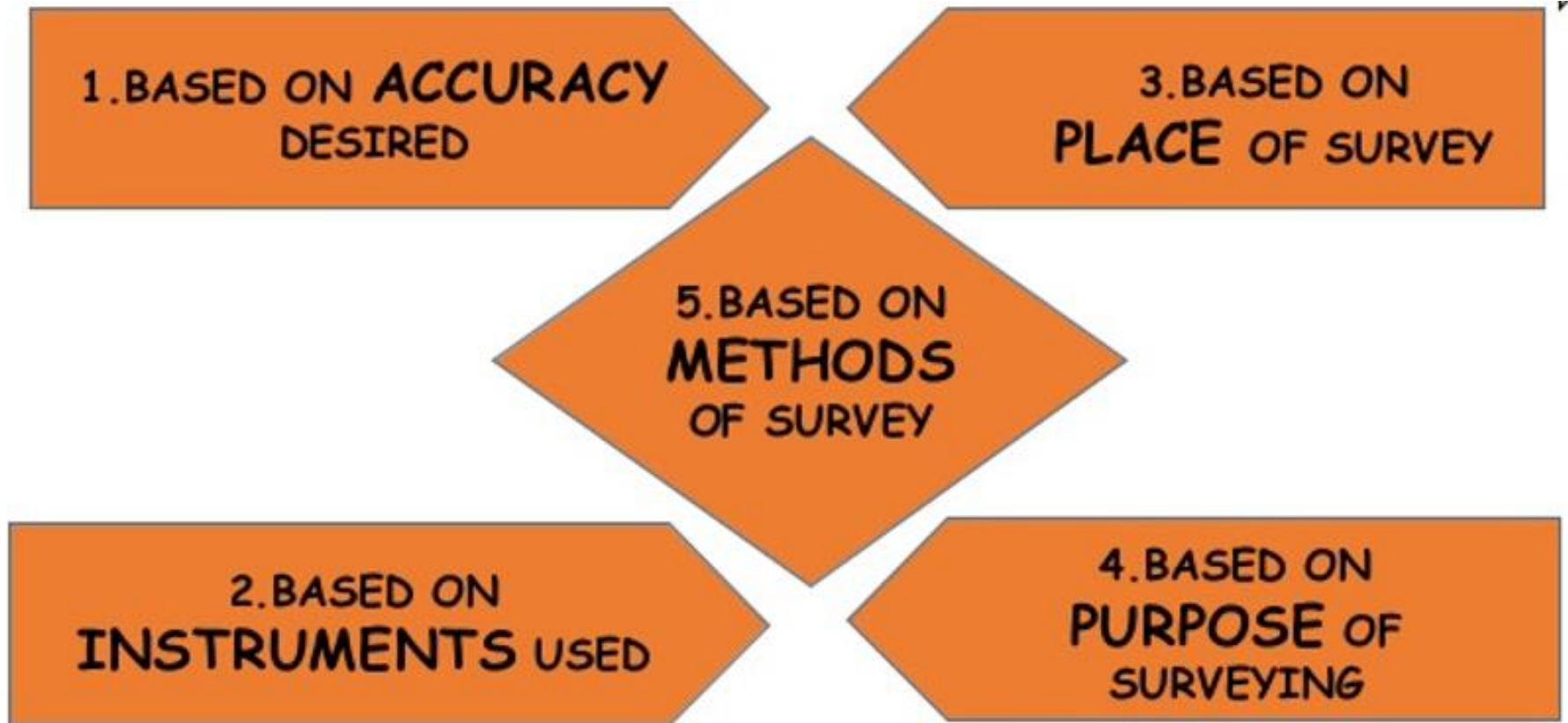
Locate the position of a point by measurement from two reference points



According to the second principle the points are located by linear or angular measurement or by both in surveying.

- (a) Taking linear measurement from A and B for C.
- (b) Taking linear measurement of perpendicular from D to C.
- (c) Taking one linear measurement from B and one angular measurement as angle ABC
- (d) Taking two angular measurement at A & B as angles CAB and ABC.
- (e) Taking one angle at B as $\angle ABC$ and one linear measurement from A as AC.

Classification of Surveying

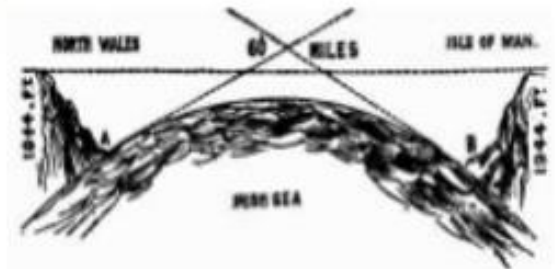


Classification of Surveying

BASED ON ACCURACY DESIRED:

PLANE SURVEYING

- Type of surveying in which earth surface is considered as a plane and the curvature of the earth is ignored.
- In such surveying the line joining any two stations is considered to be straight.



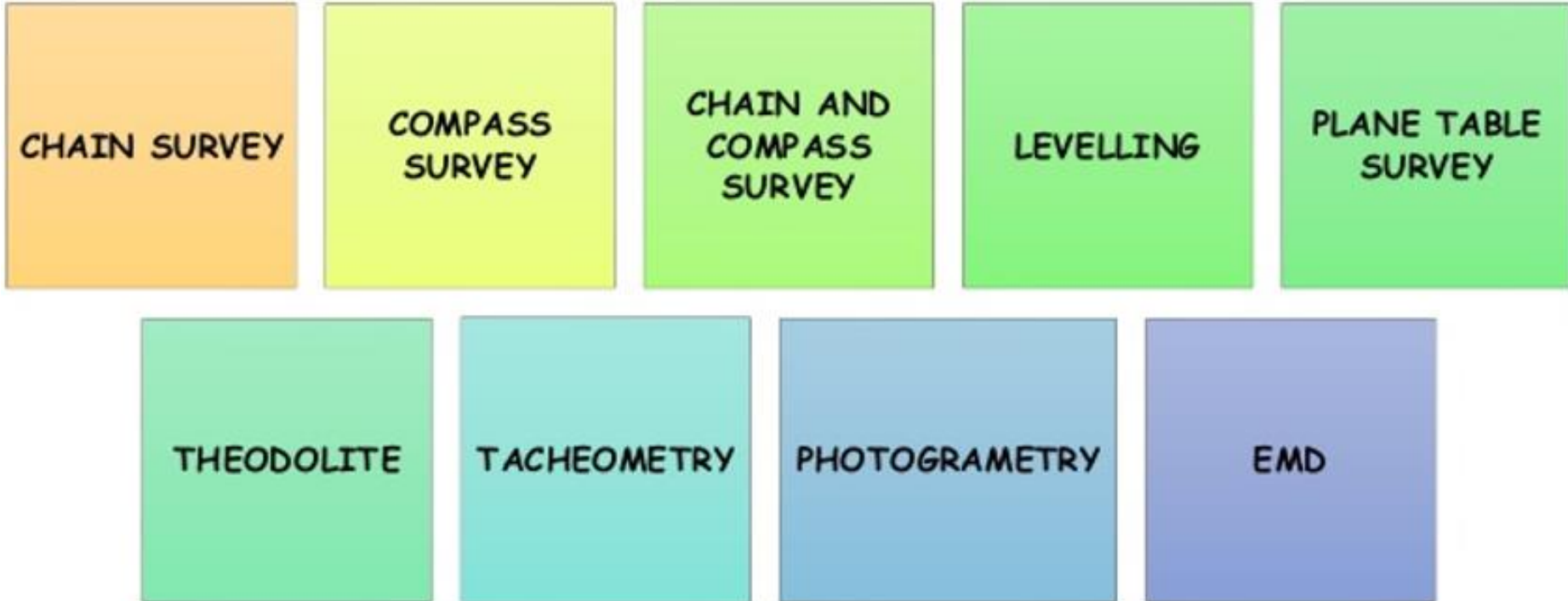
GEODETIC SURVEYING

- Type of surveying in which the curvature of the earth is taken into consideration.
- The line joining any two stations is considered as curved line.



Classification of Surveying

BASED ON INSTRUMENTS USED:

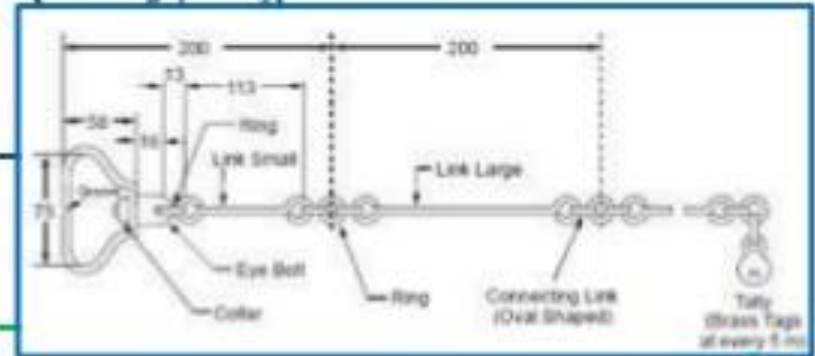
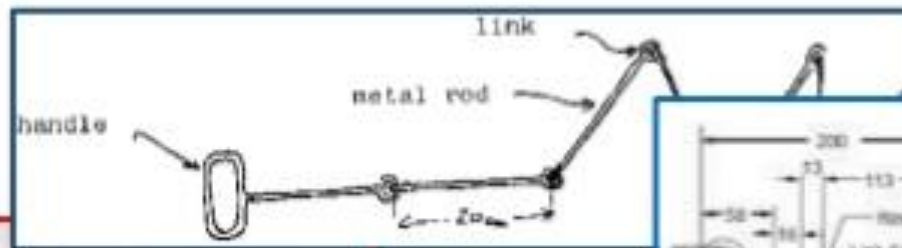


Classification of Surveying

BASED ON INSTRUMENTS USED:

CHAIN SURVEY

- Simplest type of surveying in which only linear measurements are made with a chain or tape. **Angular measurements are not taken.**

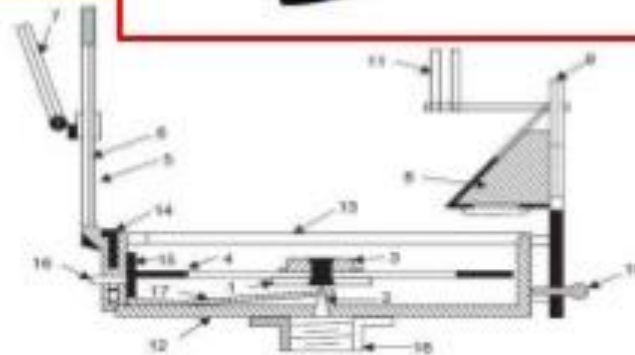


Classification of Surveying

BASED ON INSTRUMENTS USED:

COMPASS SURVEY

- ❑ The angles are measured with the help of a magnetic compass.



- | | |
|------------------------------|------------------------------|
| 1. Needle | 10. Focussing shield |
| 2. Pivot | 11. Dark sunglasses |
| 3. Agate cap | 12. Box |
| 4. Graduated disc | 13. Glass cover |
| 5. Old metal frame | 14. Lifting pin |
| 6. Horse hair | 15. Light spring |
| 7. Mirror | 16. Slide pin or knob |
| 8. Reflecting prism with cap | 17. Lifting lever |
| 9. Eye vane | 18. Support for 11 on tripod |

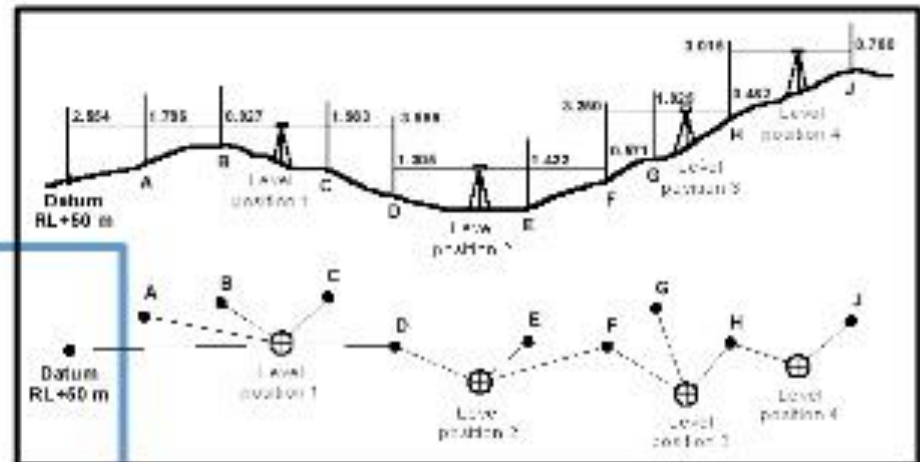
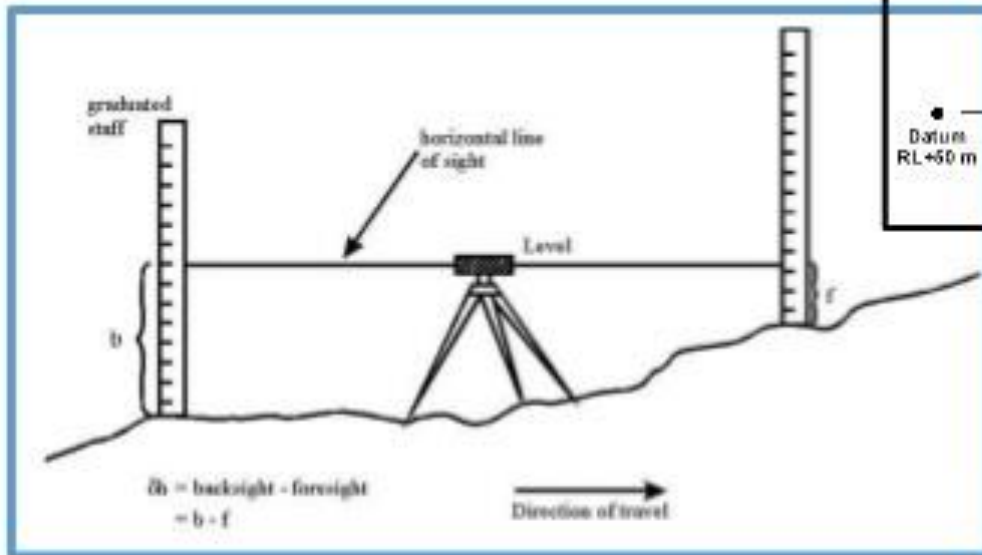


Classification of Surveying

BASED ON INSTRUMENTS USED:

LEVELLING SURVEY

- The type of survey is used to determine the vertical distances (elevation) and relative heights of points with the help of an instrument known as level.

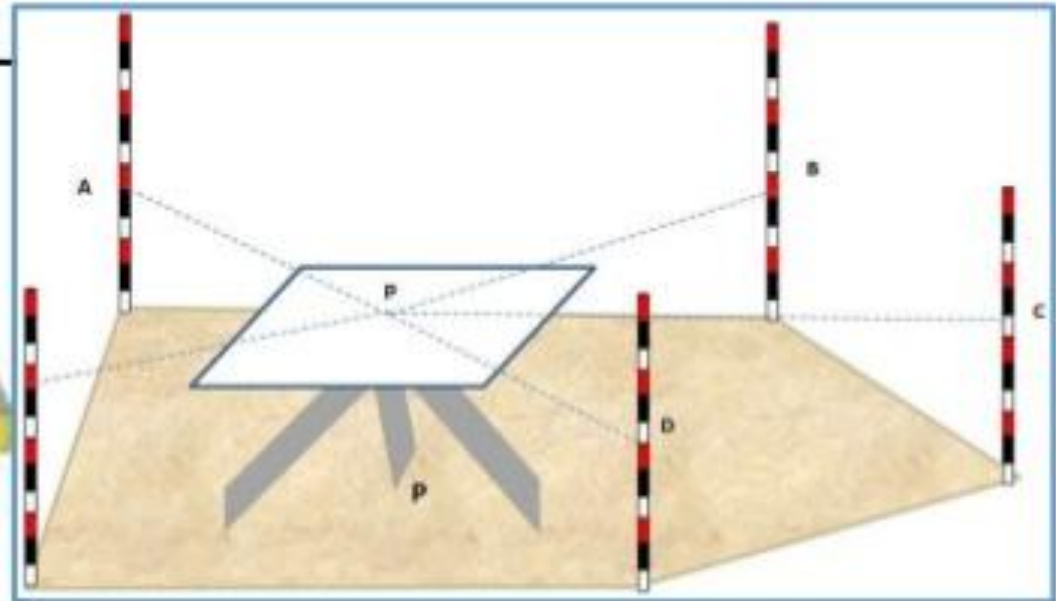


Classification of Surveying

BASED ON INSTRUMENTS USED:

PLANE TABLE SURVEY

- ❑ It's a graphical method of surveying in which field works and plotting both are done simultaneously.



Classification of Surveying

BASED ON INSTRUMENTS USED:

THEODOLITE SURVEY

- ❑ THEODOLITE Is a precision instrument for measuring angles in the horizontal and vertical planes.

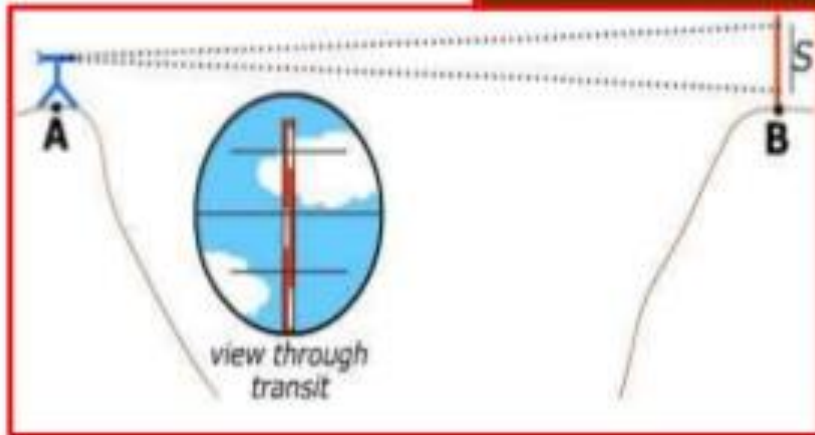


Classification of Surveying

BASED ON INSTRUMENTS USED:

TACHEOMETRIC SURVEY

- is a system of rapid surveying, by which the positions, both horizontal and vertical, of points on the earth surface relatively to one another are determined without using a chain or tape or a separate leveling instrument.

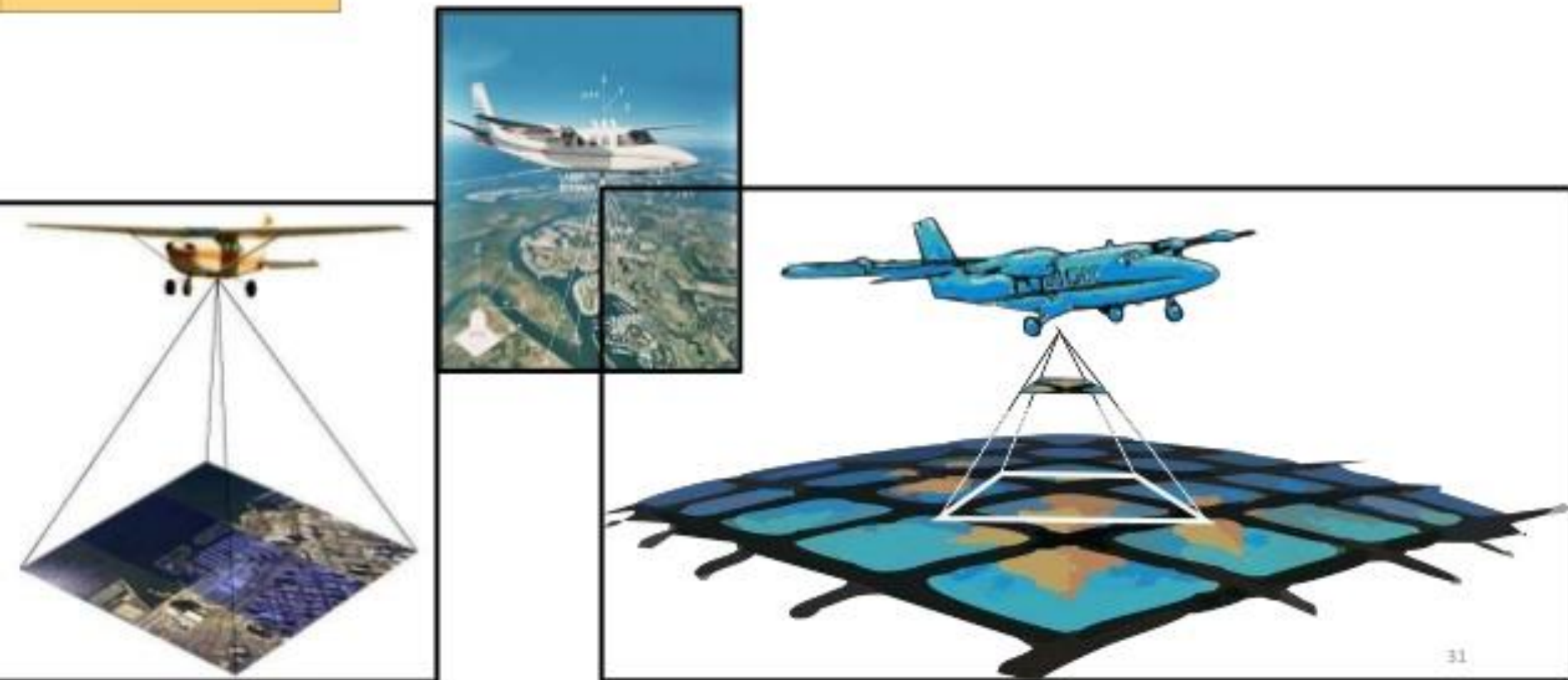


Classification of Surveying

BASED ON INSTRUMENTS USED:

PHOTOGRAMMETRY SURVEY

- ❑ Photogrammetry is the science of making measurements from photographs, especially for 3D data acquisition and object reconstruction

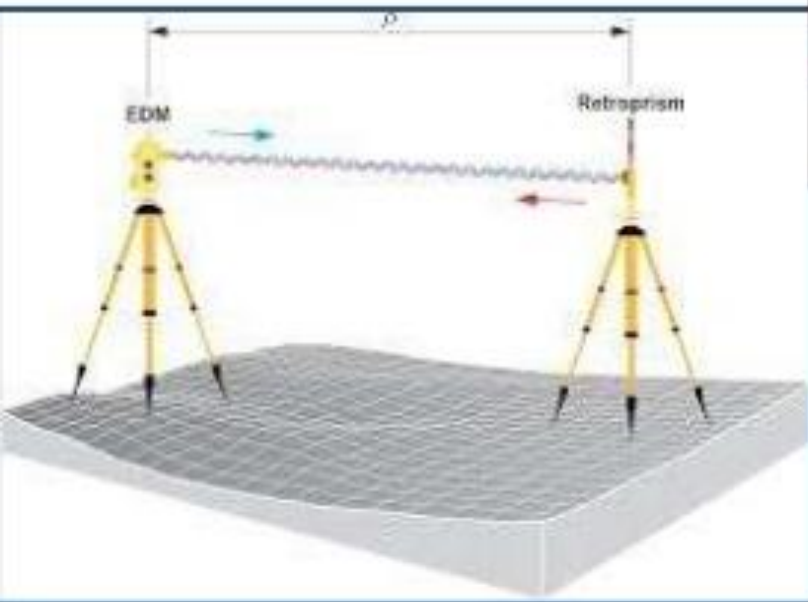


Classification of Surveying

BASED ON INSTRUMENTS USED:

EDM SURVEY

- ❑ Electronic distance measuring instrument is a surveying instrument for measuring distance electronically between two points through electromagnetic waves.

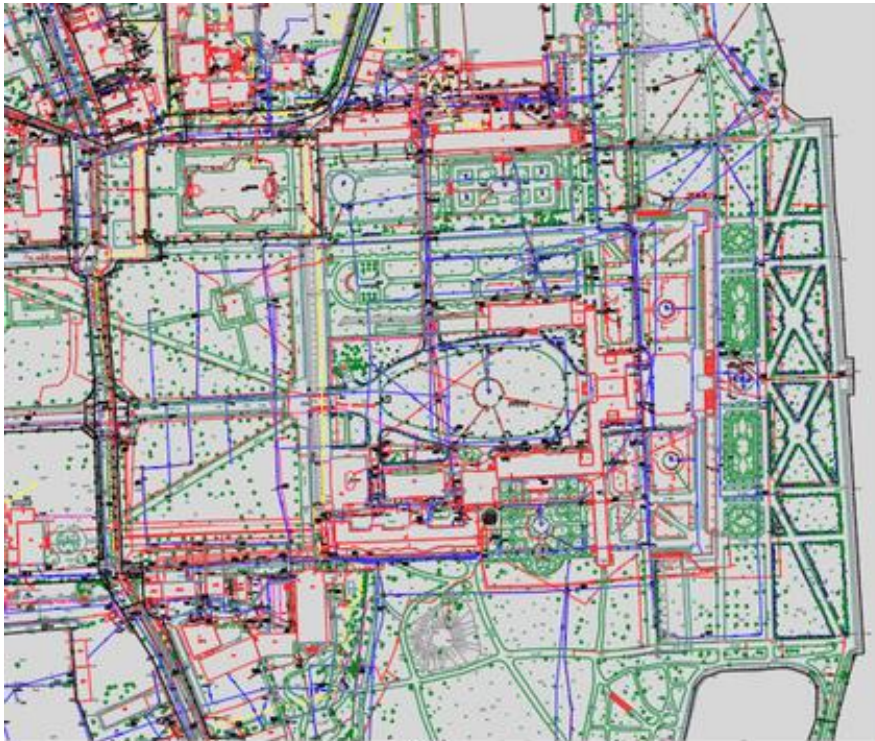


Classification of Surveying

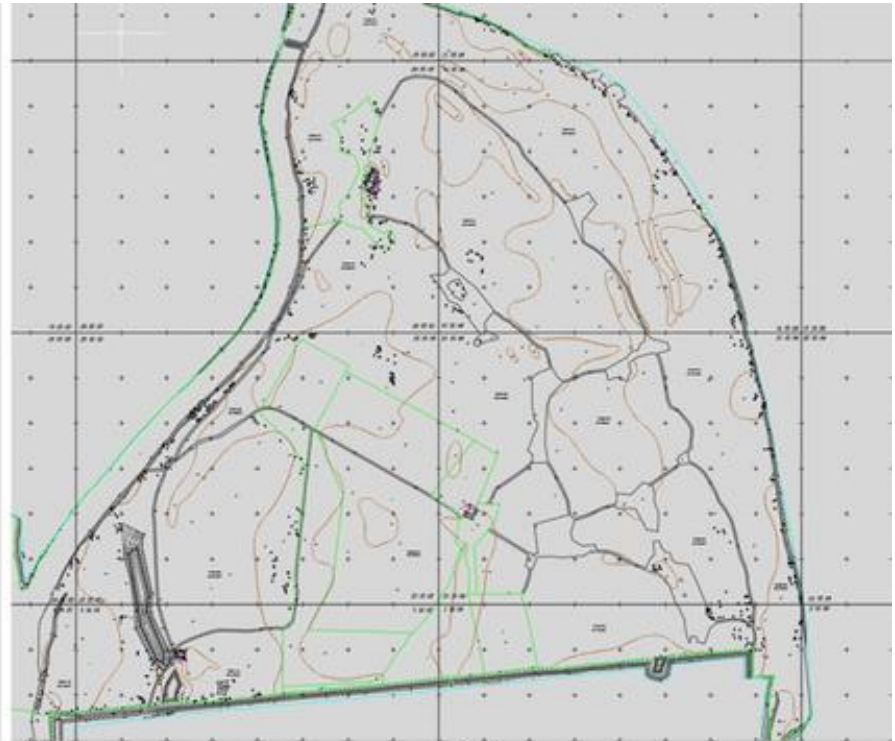
BASED ON PLACE OF SURVEY:

Land Survey

Land Survey is done on land to prepare plan and maps of a given area. Topographical, city and cadastral surveys are some of the examples of land surveying.



(a)



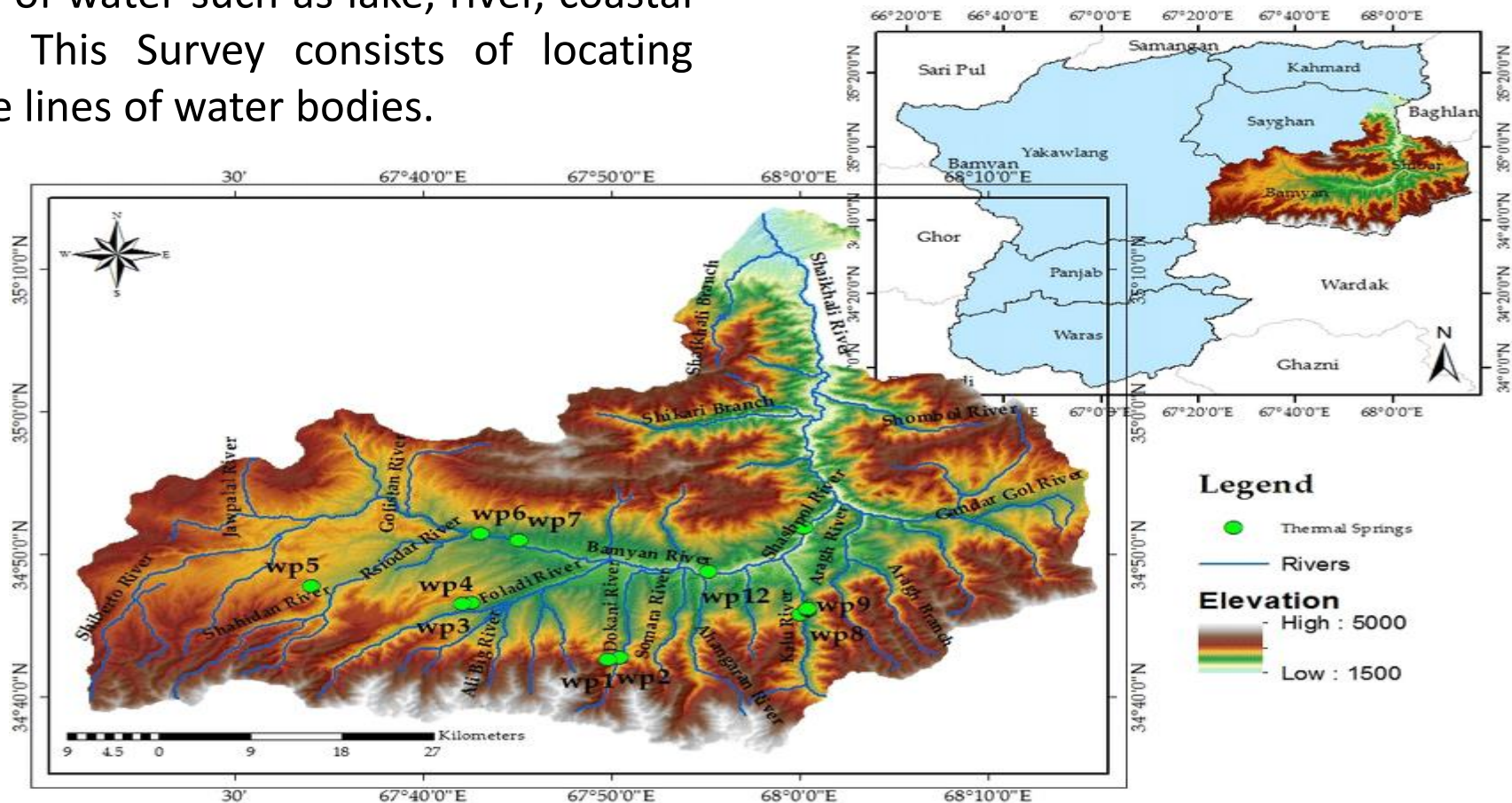
(b)

Classification of Surveying

BASED ON PLACE OF SURVEY:

Hydrological Survey

This survey is conducted on or near the body of water such as lake, river, coastal area. This Survey consists of locating shore lines of water bodies.

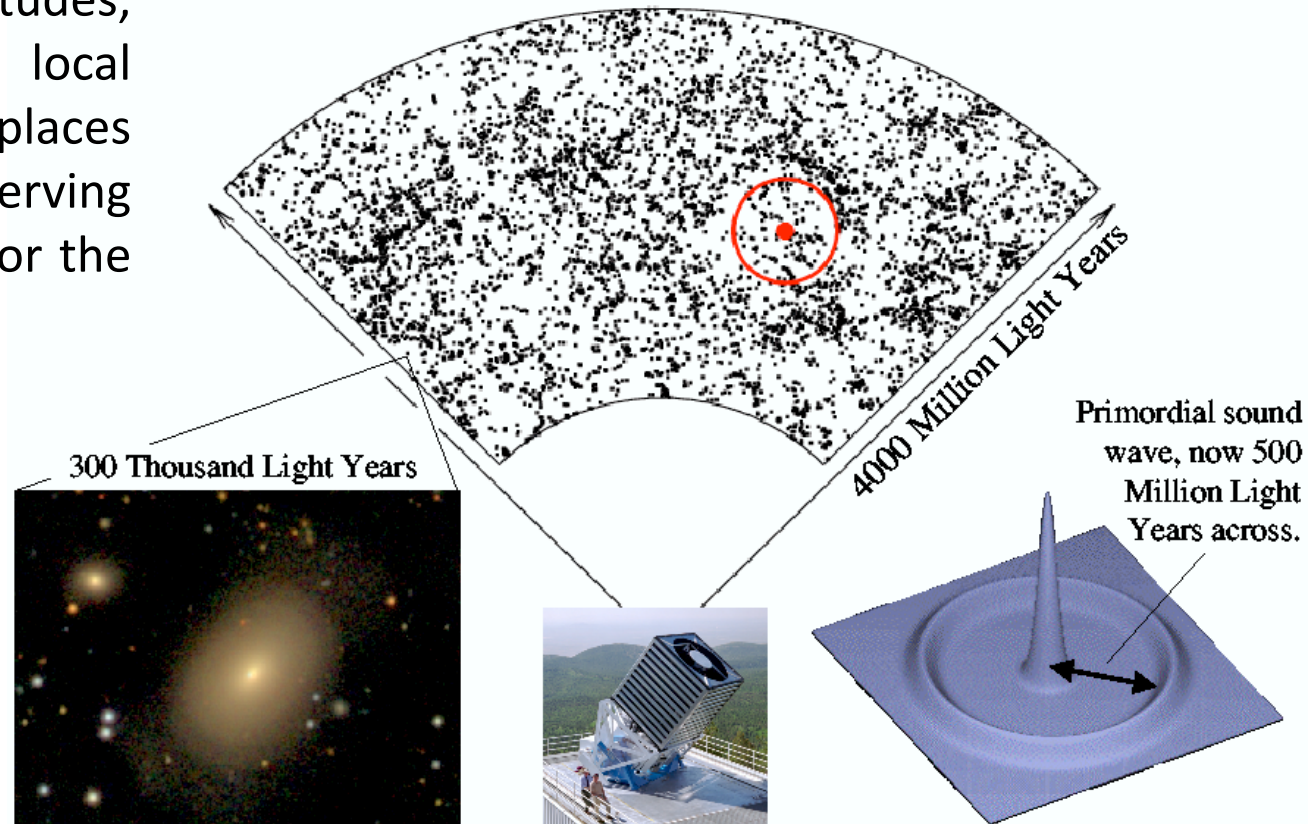


Classification of Surveying

BASED ON PLACE OF SURVEY:

Astronomical Survey

This survey is conducted for the determining of latitudes, longitudes, azimuths, local time, etc. for various places on earth by observing heavenly bodies (sun or the stars).



Classification of Surveying

BASED ON PLACE OF SURVEY:

Aerial Survey

An aerial survey is conducted from aircraft. Aerial cameras take photographs of the surface of the earth in overlapping strips of land. This is also known as photographic survey.

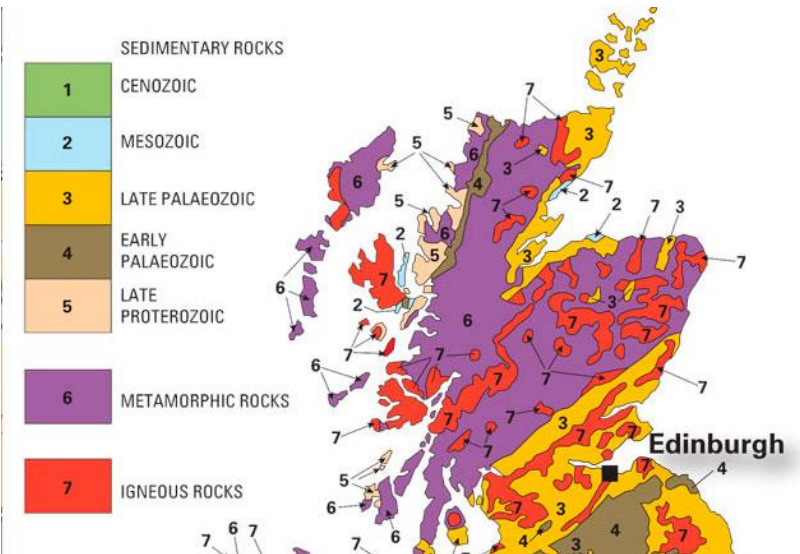


Classification of Surveying

BASED ON PURPOSE OF SURVEY:

Geological Survey

In this both surface and subsurface surveying are conducted to locate different minerals and rocks. In addition, geological features of the terrain such as folds and faults are located.

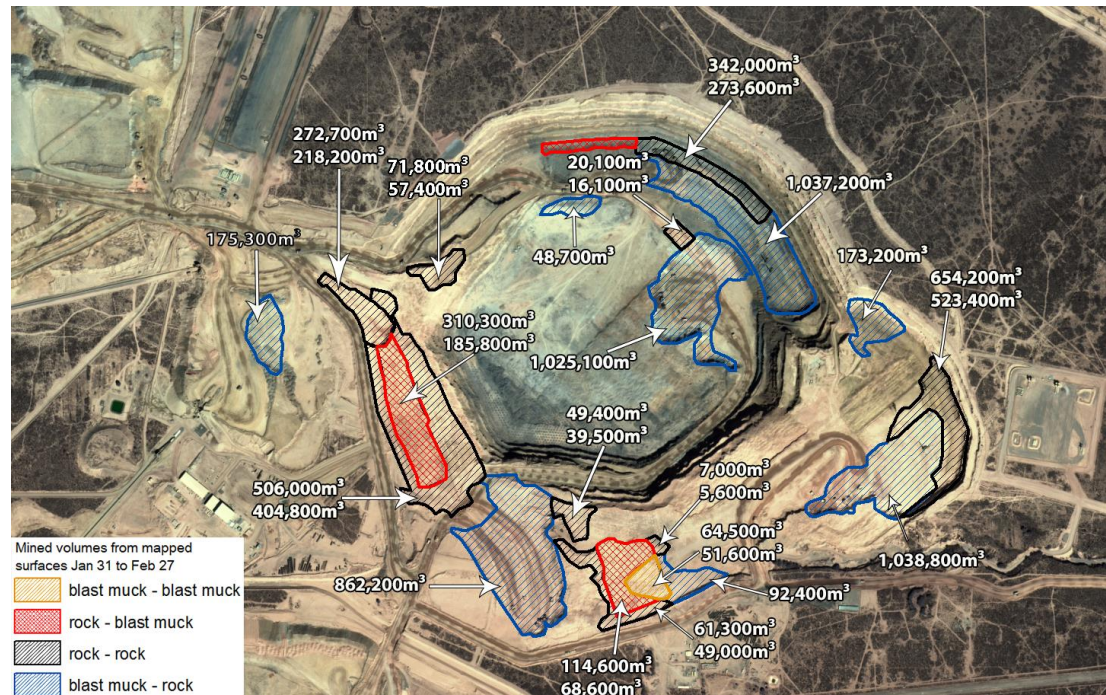


Classification of Surveying

BASED ON PURPOSE OF SURVEY:

Mine Survey

Mine Survey includes include both surface and underground surveys. It is conducted for the exploration of mineral deposits and to guide tunneling and other operations associated with mining.

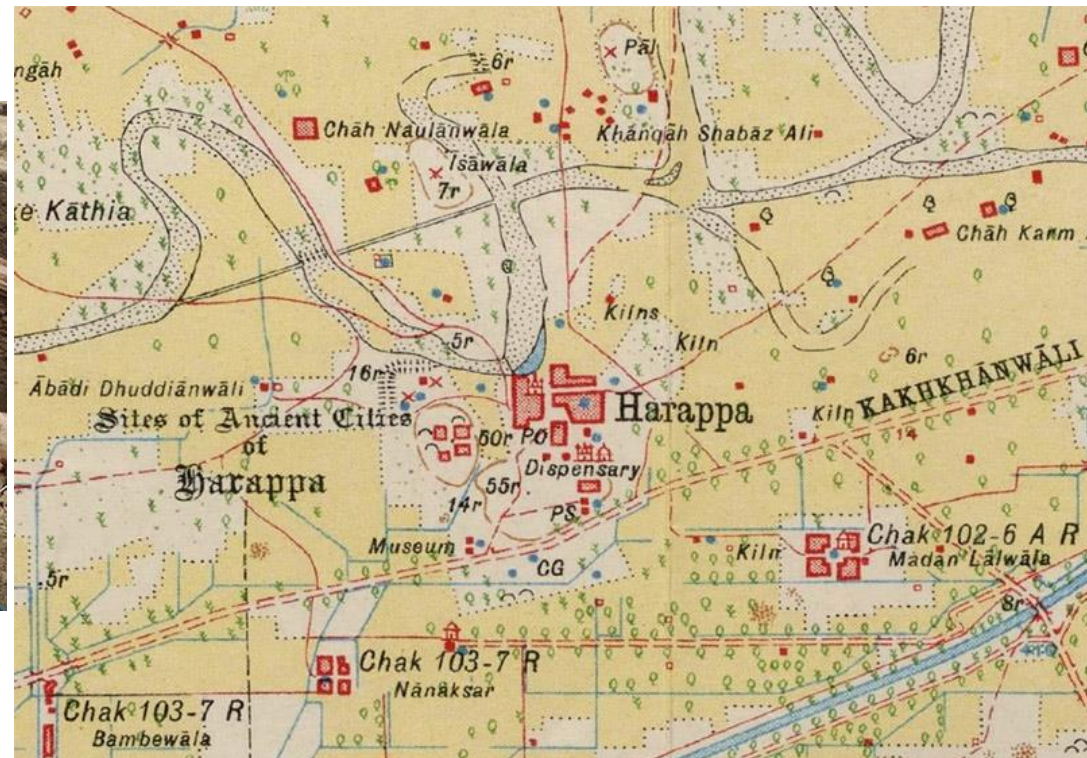


Classification of Surveying

BASED ON PURPOSE OF SURVEY:

Archaeological Survey

It is conducted to locate relics of antiquity, civilization, kingdoms, forts, temples, etc.



Classification of Surveying

BASED ON PURPOSE OF SURVEY:

Military Survey

It has a very important and critical applications in the military. Aerial surveys are conducted for this purpose. It is conducted to locate strategic positions for the purpose of army operations.

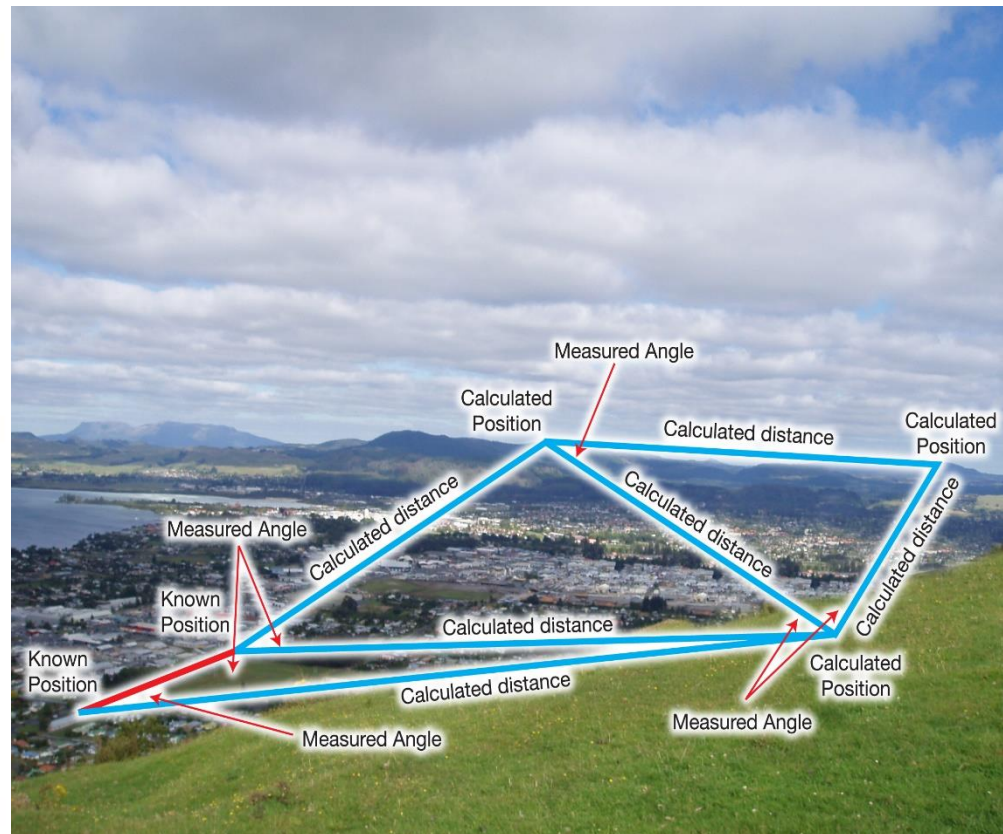
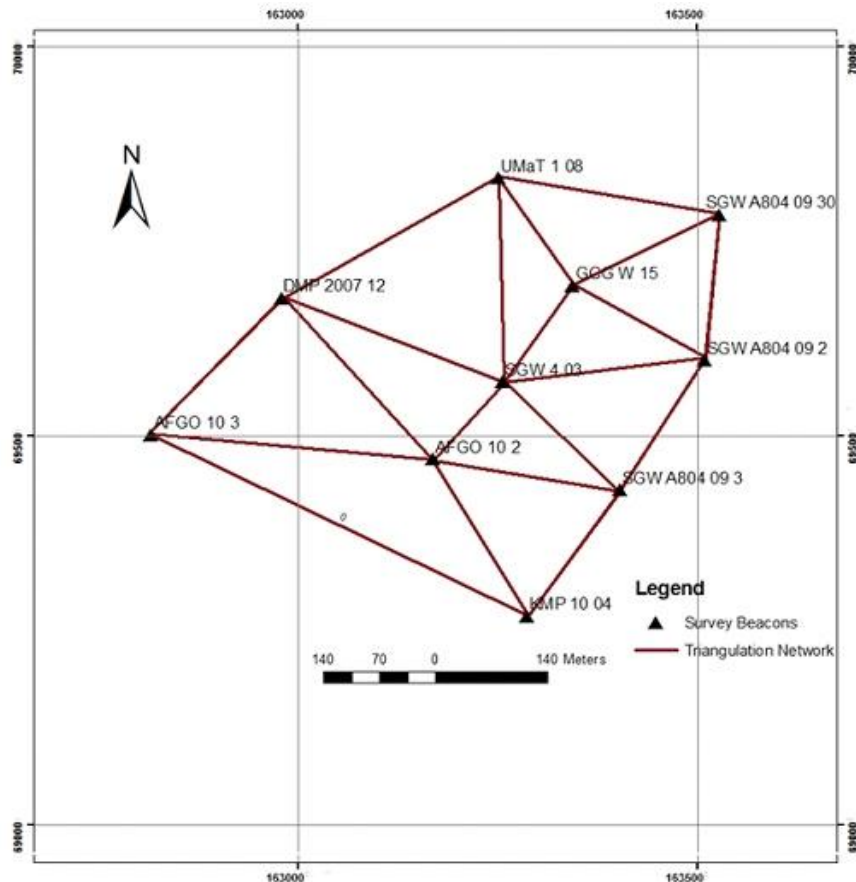


Classification of Surveying

BASED ON METHOD OF SURVEY:

Triangulation

Triangulation is basic method of surveying, when the area to be surveyed is large, triangulation is adopted. The entire area is divided into network of triangles.

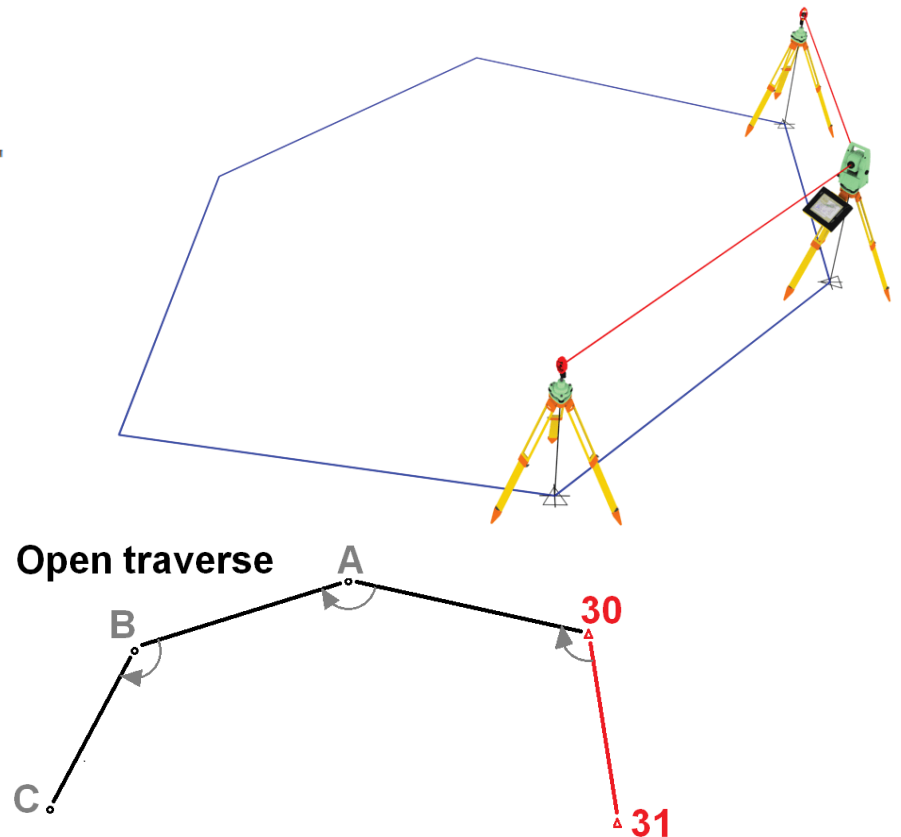
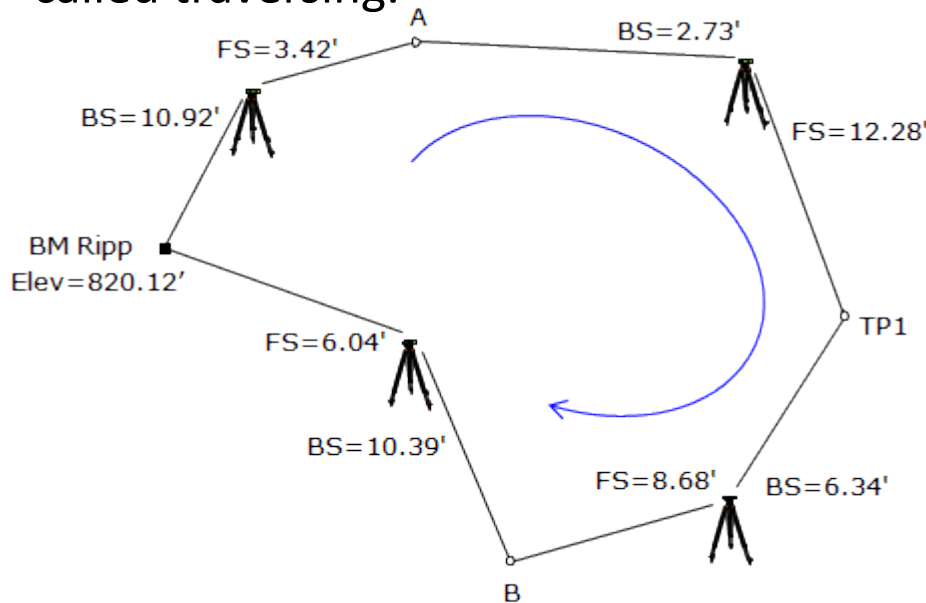


Classification of Surveying

BASED ON METHOD OF SURVEY:

Traversing

A Traversing is circuit of survey lines. It may be open or closed. When the linear measurements are done with a chain and a tape and the directions or horizontal angles are measured with a compass or a theodolite respectively the survey is called traversing.



Plan and Maps

DIFFERENCE BETWEEN PLAN AND MAP:

(1) **Plan:** Plan is a graphical representation to some scale of the features on near or below the surface of the earth as projected on a horizontal plane. ✓

(2) **Map:** If the graphical projection on a horizontal plane is small, the plan is called a map.

Map	Plan
1. We can study a part or whole of the earth with the help of a map. 2. A map contains a lot of information. 3. A map shows only the very important features of the area.	1. A plan is a detail of drawing of small areas. 2. In plan, details are given in the form of symbols. 3. A plan can show the length and the breadth.

Plan and Maps

DIFFERENCE BETWEEN PLAN AND MAP:

MAP



PLAN



Scales and Units of Measurements

SCALES AND ITS TYPES

A scale is the ratio of the distance marked on the plan to the corresponding distance on the ground. A good draughtsman can plot a length to accuracy within 0.25 mm.

Types of Scales are generally classified as large, medium and small.

- Large scale : 1 cm = 10 m or less than 10 m.
- Medium scale : 1 cm = 10 m to 100 m.
- Small scale : 1 cm = 100 m or more than 100 m.

REPRESENTATION OF A SCALE

- **Engineer's scale** : 1 cm = 50 m
- **Representative Fraction (R.F)** : It is the ratio of the distance on the map to the corresponding distance on the ground taken as same units. Scale of 1 cm = 50 m, 1 cm on the map represents 50 m (5000 cm) on the ground. Therefore, the representative fraction (R.F.) is $1/5000$ or 1: 5000.

Scales and Units of Measurements

REPRESENTATION OF A SCALE

- Graphical scale** : A graphical scale is a line drawn on the map so that its map distance corresponds to a convenient units of length on the ground. It has the advantage over the numerical scales that the distances on the maps can be determined by actual scaling even when the map has shrunk.

Purpose of Survey	Scale	R.F
Building Site	1 cm = 10 m	1:1000
Town Planning, Reservoir planning, etc	1 cm = 50 cm to 100 m	1:5000 to 10000
Route Surveys	1 cm = 10 m to 60 m	1:1000 to 1:6000
Longitudinal Sections.	1 cm = 10 m	1:1000
	1 cm = 1 m	1:100
Cross- Sections	1 cm = 1 m	1:100
Land Surveys/ Cadastral Surveys	1 cm = 10 m to 50 m	1:1000 to 1:5000
Topographical Maps	1 cm = 0.25 km to 2.5 km	1:25000 to 1:250000
Geographical Maps	1 cm = 5 km to 150 km	1:500000 to 1:15000000
Mine Surveys	1 cm = 10 m to 25 m	1:1000 to 1:2500
Forest Maps	1 cm = 250 m	1:25000

Scales and Units of Measurements

UNITS OF MEASUREMENTS

According to Standards of Weights and Measurements Act, India decided to give up FPS system used earlier and switched over to MKS in 1956. In 1960 System International (SI units) unit was approved by the conference of weights and measures. It is an international organisation of which most of the countries are the members. In this system also unit of linear measurement is metre. However, in this system use of centimeters and decameters are discouraged. Of course major difference between MKS and SI is in the use of unit of force. In MKS unit of force is kg-wt (which is commonly called as kg only) while in SI it is newton.

The recommended multipliers in SI units are given below

Giga unit = 1×10^9 units

Mega unit = 1×10^6 units

Kilo unit = 1×10^3 units

unit = 1×10^0 units

Milli unit = 1×10^{-3} unit

Micro unit = 1×10^{-6} unit

Commonly used linear units in surveying are kilometre, metre and millimetres. However centimetre is not yet fully given up.

For measuring angles sexagesimal system is used. In this system:

1 circumference = 360°

1 degree = $60'$ (minutes of arc)

1 minute = $60''$ (seconds of arc)

Linear Measurements

DEFINITION AND TYPES

In engineering survey, the linear horizontal distance is to be measured to complete a survey. The measurement of this linear horizontal distance between two points on the earth surface is known as linear measurement. Linear measurement can be achieved by various instruments. The measurement type can be divided as

- **Direct measurement methods (using chain, tapes)**
- **Optical measurement methods (using tacheometer)**
- **Electronic distance measurement method**

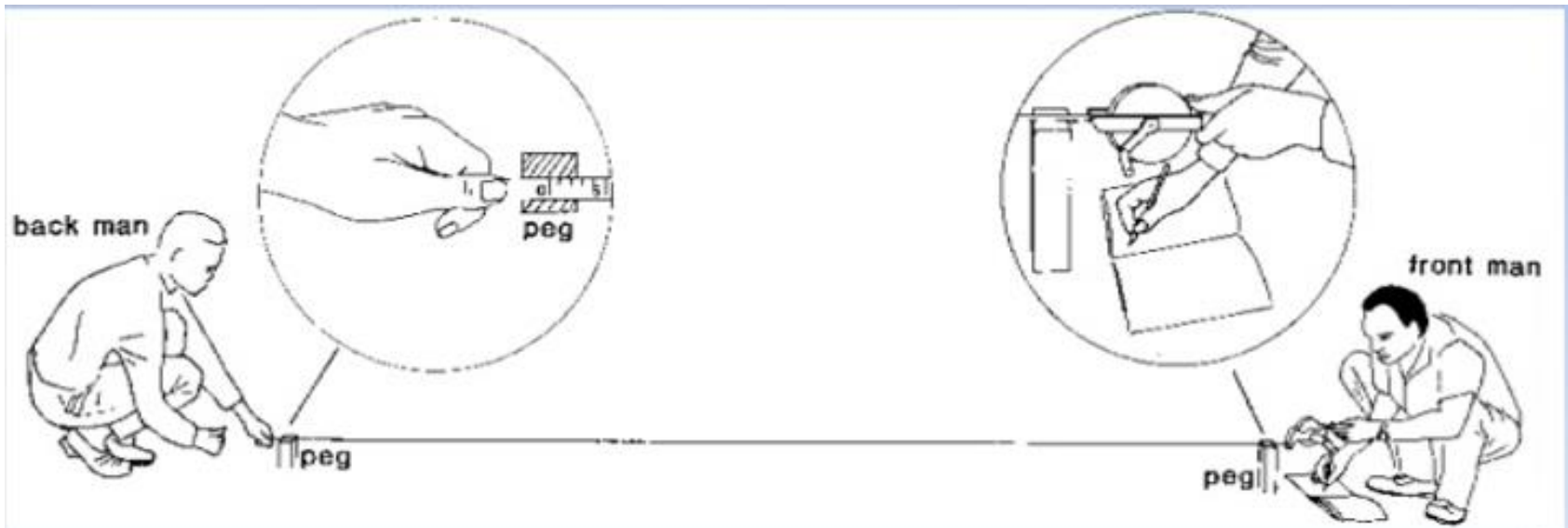


Linear Measurements using tapes

MEASURING TAPES

Tapes are used for more accurate measurements. Tapes are classified based on the materials of which they are made such as:

- Cloth or linen tape
- Fibre tape
- Metallic tape
- Steel tape
- Invar tape



Linear Measurements using tapes

TYPES OF MEASURING TAPES

CLOTH OR LINEN TAPE

- made of closely woven linen
- width of the strip is about 12 to 16 mm
- available in different lengths such as 10m, 20m, 30m, and 50m
- light in weight and easy to handle.
- may shrink when exposed to water and also elongate when pulled.
- not suitable for accurate surveying measurements.
- generally used for measuring offsets and for ordinary works.



Linear Measurements using tapes

TYPES OF MEASURING TAPES

METALLIC TAPE

- improved version of linen tape.
- Brass or copper made wires are used as reinforcement for the linen material.
- more durable than normal linen tape.
- available in different lengths of 2m, 10m, 15m, 20m, 30m, and 50m.
- used for survey works such as topographical survey works where minor errors are not taken into consideration.



Linear Measurements using tapes

TYPES OF MEASURING TAPES

FIBRE TAPE

- made of glass fibers coated with PVC.
- light in weight and flexible.
- available in lengths of 5m, 10m, 20m, 30m, and 50m
- may stretch when subjected to tension. Hence, these are not suitable for accurate surveying works.
- are recommended in place of steel tapes where it is essential to take measurements in the vicinity of electric fences and railway lines, etc.



Linear Measurements using tapes

TYPES OF MEASURING TAPES

STEEL TAPE

- made of steel or stainless steel.
- consists of a steel strip of 6mm to 16mm wide.
- available in lengths of 1m, 5m, 8m, 10m, 20m, 30m and 50m.
- Meters, decimeters, and centimeters are graduated in the steel strip.
- The tape is withdrawn from the case by using a hand during measuring and it is rewound into the case by just pressing button provided on the case.
- not flexible and are suitable for measuring leveled surfaces only.
- may corrode easily when exposed to moisture and to prevent this tape, it should be cleaned and oiled after every use.
- generally used for standardizing chains, measurements of construction works, etc.



Linear Measurements using tapes

TYPES OF MEASURING TAPES

INVAR TAPE

- made of an alloy which consists of 36% of nickel and 64% of steel.
- contains a 6mm wide strip and is available in different lengths of 30m, 50m, 100m.
- The coefficient of thermal expansion of invar alloy is very low hence not affected by changes in temperature.
- used for high precision works in surveying such as baseline measurement, triangulation surveys, etc.
- expensive than all the other types of tapes.
- should be handled with care otherwise bends or kinks may be formed.



Ranging in Surveying

DEFINITION

Ranging is the process of establishing many intermediate points to measure the survey lines in linear measurement. When the length of the surveying line is longer than the length of the measuring chain, the line can be measured by using intermediate points along with it by ranging.

INSTRUMENTS USED

Ranging can be done by using the following instruments:

- Ranging rods
- Offset rods
- Ranging poles

Ranging rods are vertical rods used to mark survey points having 2-3 m height. It is painted with bands of red and white 20 cm each.

Offset rods are used for measuring offset of short lengths. It is similar to ranging rod and is usually of 3m lengths.

Ranging poles are used for spotting and marking the position of survey stations. If a range rod is longer than 12 feet, it's called a ranging pole.



Types of Ranging

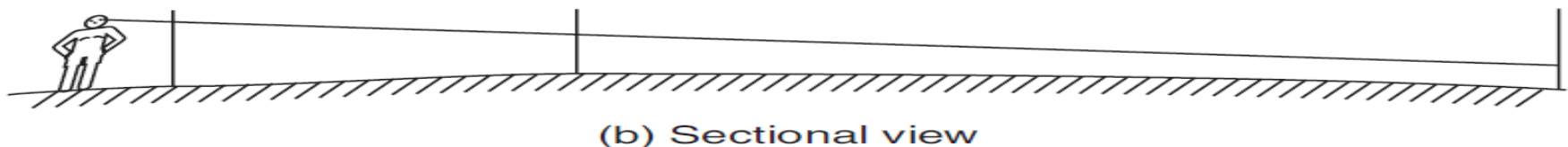
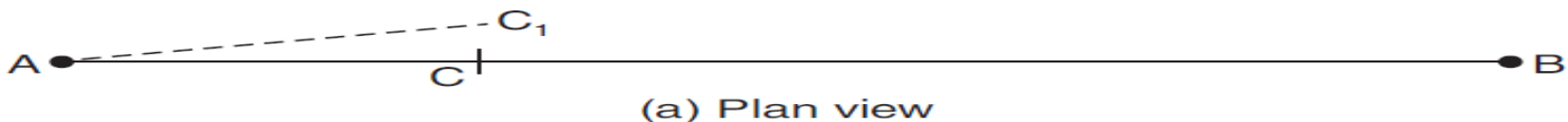
Ranging can be of two types

1. Direct ranging
2. Indirect ranging

Direct Ranging

Direct ranging is adopted when the two survey stations are inter-visible i.e. the two end points of the survey line are visible..

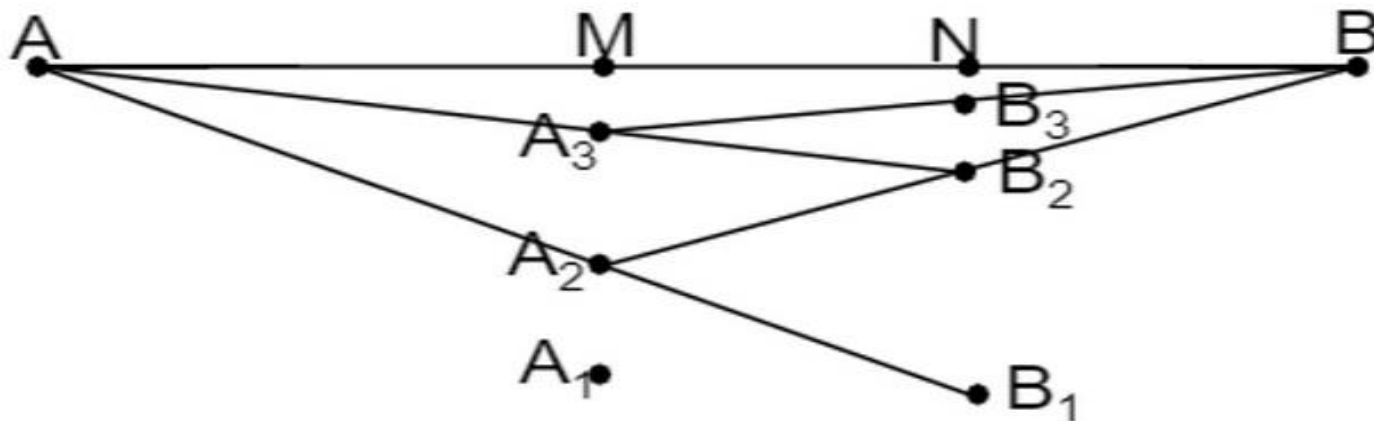
For example, A and B are the two inter-visible points at the survey end line. At points A & B ranging rods are fixed. Then, another person takes the ranging rod and fixes or establishes an intermediate point C between the line AB. The surveyor at point A then signals another person at the intermediate points so that the ranging rod is in perfect line with the endpoints A and B. Hence, the intermediate points are determined.



Types of Ranging

Indirect Ranging

- It is used when the two endpoints of the survey line are either not inter-visible or the two points are at a very long distance. For instance, let A and B be the points that are not visible. Now following steps are carried out:
 - A1 and B1 are considered such that A and B1 are visible from A1 and A1 and B are visible from B1.
 - A1 is shifted to a position A2 such that A, A2, and B1 are in the same line.
 - Similarly, B1 is shifted to B2 such that A2, B2, and B are in the same line.
 - The process is repeated until all the points are in the same line and hence intermediate points are located.



Obstacle to Ranging

These obstacles can be further classified into the following categories:

(a) Both ends of the line are visible from some intermediate points. Intervening ground is an example of such obstacle. By resorting to reciprocal ranging this difficulty can be overcome.

(b) Both ends of the line may not be visible from intermediate points on the line, but may be visible from a point slightly away from the line. Intervening trees and bushes are the examples of such obstacles. This obstacle to chaining may be overcome by measuring along a random line as shown in Fig. In this case required length.

$$EB = \sqrt{EC^2 + CB^2}$$

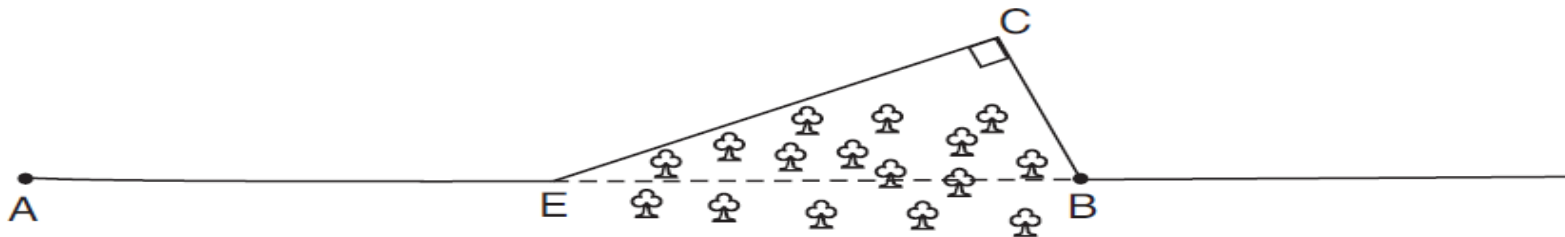


Fig. 12.20. Obstacle to ranging

Angular Measurement

- Points on ground or on map are related to each other through horizontal distance and **horizontal angle (direction)**.
- Horizontal angular measurement are made between survey lines to determine angle between the lines.
- Horizontal angle is difference between two measured directions.

When the area to be surveyed is large, angles or direction is essential to be observed while drawing survey lines. Such angular measurements are done using **COMPASS**.

Types of Compass



Prismatic Compass



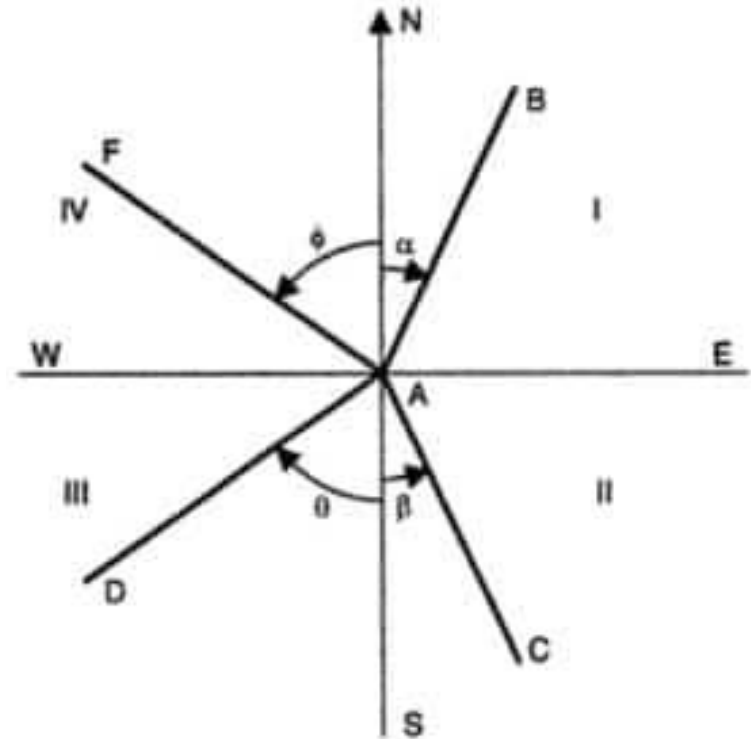
Surveyor Compass

Angular Measurement

Bearing

- Bearing is defined as the acute angle that is measured between the reference meridian and the given line.
- The line is measured either from the north or south towards east or west that would give an angle less than 360 degrees. The angle is represented by N or S first followed by angle value and E or W direction, For example, N60E.

From the figure, the bearing of line AB is ? which is less than 90 degrees as it lies in first quadrant NE.



Theodolite

Theodolite is a measurement instrument utilized in surveying to determine horizontal and vertical angles with the tiny low telescope that may move within the horizontal and vertical planes.

Followings are the major uses of theodolite:

- Measuring horizontal and vertical angles
- Locating points on a line
- Finding the difference in the level
- Prolonging survey lines
- Ranging curves
- Setting out grades



Total Station

A Total Station is a modern surveying instrument that integrates an **electronic theodolite with an electronic distance meter**.

Total Stations use electronic transit theodolites in conjunction with a distance meter to read any slope distance from the instrument to any particular spot. They are hence two essential surveying instruments in one and when used with other technology such as mapping software are able to deliver the 'total' surveying package, from measuring to mapping.



Laser Distance Measurement

- Laser tape measures, also known as laser distance measures or laser distance meters, are a modern alternative to the traditional tape measure.
- Measurements are taken by sending out a pulse of laser light to a defined target and then measuring the amount of time it takes for the reflection to return.
- This allows for extremely accurate readings, even over large distances.
- Additionally, on-board functions allow the device to add, subtract, calculate area and volume, triangulate and a range of other functions.
- These have a huge benefit to the user as complex, time-consuming calculations can be made quickly, easily and with high degrees of accuracy.



Global Positioning System

- The term GPS full form is “Global Positioning System” which is a satellite navigation system that provides location and time information in all climate conditions to the user.
- GPS is used for navigation in planes, ships, cars, and trucks also.
- GPS provides continuous real-time, 3-dimensional positioning, navigation, and timing worldwide.

Working of GPS

GPS uses a network of satellites, which communicate with receivers on the ground. When a receiver requests data to calculate its location, four or more GPS satellites will communicate with the receiver, sending the position of the satellite, the time the data was transmitted and the distance between the satellite and the receiver. The information collected from these satellites then calculates the **latitude, longitude and height** of the receiver. If the receiver is moving, continuous data collection can be used to calculate the changing position of the receiver over time, which can be used to calculate speed.

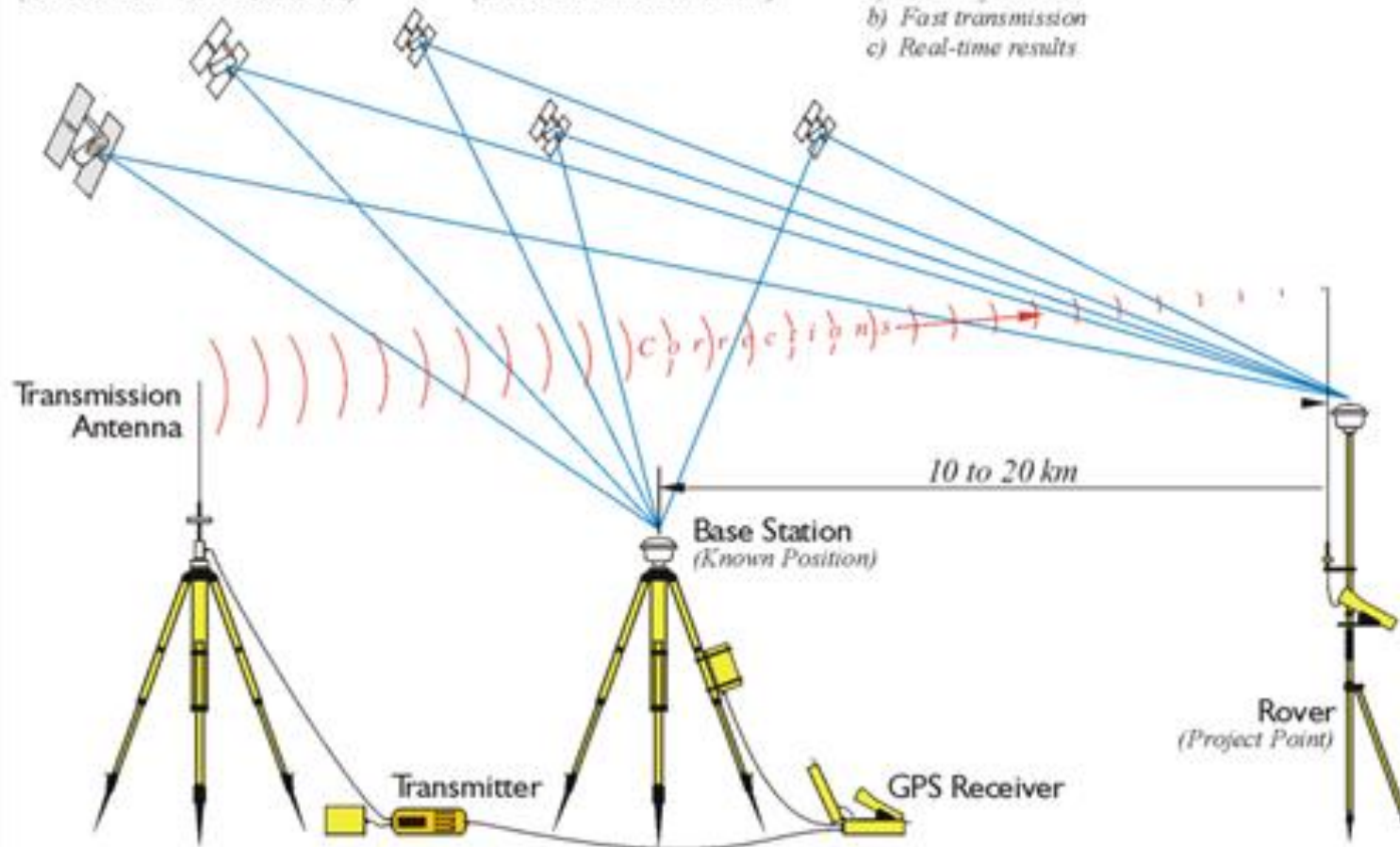
Global Positioning System

Working of GPS

Real-Time-Kinematic

Positional Accuracy ± 2 cm or so

- Same Satellite Constellation
(Base Station - Rover/or Rovers)
- Carrier Phase
(Track 5 Satellites Minimum)
- Radio Link
 - a) More information
 - b) Fast transmission
 - c) Real-time results



Geographical Information System

- GIS is short for Geographic Information System. It is a collection of computer software and information which is used for viewing, analysing, managing and displaying geographical data.
- In general, it allows users to search for information about specific geographical areas, analyse spatial information, edit the data and create maps, charts and reports that show users the results in visual forms.

Uses

- **Environmental Geography** – to analyse the impact people have on the environment.
- **Physical Geography** – to study the elements of atmosphere, biosphere and geosphere.
- **Emergency Management Information System** – to give real time data to emergency responders about the geographical layout.
- **Health Geography** – to use geographical information to study health related issues such as disease and illness.
- **Economic Geography** – to study economic activities across the earth.
- **Transportation Geography** – to investigate the spatial interactions of people or things.

Remote Sensing

Remote sensing is the science of acquiring information about the Earth's surface without actually being in contact with it.

This is done by sensing and recording reflected or emitted energy and processing, analyzing, and applying that information.

Uses of Remote Sensing

- ❧ the geographer, who looks for changes on the Earth's surface that need to be mapped;
- ❧ the forester, who needs information about what type of trees are growing and if they have been affected by disease or fire;
- ❧ the environmentalist, who wants to detect, identify and follow the movement of pollutants such as oil slicks on the ocean;

Remote Sensing

Uses of Remote Sensing

- ❧ the farmer, who wants to keep an eye on how his crops are growing and if they've been affected by drought, floods, disease or pests;
- ❧ the ship captain, who needs to find the best route through the northern ice packs;
- ❧ the firefighter, who sends out his crews based on information about the size and movement of a forest fire.
- ❧ the Urban Planner, who wants to map and monitor land cover, land use, morphology (the study of the form of human settlements and the process of their formation and transformation) etc.

Remote Sensing

Process of Remote Sensing

